**Jaiden Medina:**

My name is Jaiden Medina, and I am a third-year student studying Intelligent Systems Engineering with a specialization in Cyber-Physical Systems. I am taking this course because I am strongly interested in the intersection of hardware and software, particularly in embedded systems that interact directly with the physical world. I want to deepen my understanding of how embedded systems are designed, implemented, and optimized for real-world applications.

One of the most influential experiences shaping this interest was ENGR-E 222 (Intelligent Systems II), where my team developed both cloud-based and embedded systems. My favorite project involved optimizing a Raspberry Pi–based neural network for human activity recognition in elderly patients. The system classified activities such as sitting, standing, walking, and falls, and required careful trade-offs between model size, inference speed, and accuracy. This project strengthened my interest in embedded computing and performance-constrained systems.

Through this course, I hope to build both theoretical and practical skills in embedded systems design. I am particularly excited about hands-on projects that emphasize efficiency, optimization, and real engineering constraints. I have also taken an elective course on radiation effects and reliability in microelectronics, which gave me insight into designing hardware for extreme environments. I hope to combine that perspective with the material in this course to further refine my career interests and technical direction.

**Brady Adams:**

My name is Brady Adams, and I am a third-year student studying Intelligent Systems Engineering with a specialization in Cyber-Physical Systems. I chose to take this course because of my strong interest in low-level programming and embedded systems, particularly C programming, microcontrollers, and hardware-software integration. I have developed significant programming experience both inside and outside the classroom, which I believe will be valuable in this course.

Currently, I am part of a university research project called the TechTeach program, where I work with other ISE students and education majors to develop an educational video game for middle school students. We are creating the game from scratch using C# and collaborating with a teacher client to meet classroom needs. This experience has strengthened my software development skills and reinforced my interest in applying programming to real-world systems.

My goal for this course is to further develop my technical skills in C programming while gaining hands-on experience with microcontroller-based hardware systems. I have also taken a course in Radiation Effects and Reliability in Microelectronics, which introduced me to concepts such as power management and hardware reliability. Ultimately, I hope to pursue a career in areas such as video game development or AI-driven interactive systems, and I see this course as an important step toward strengthening my embedded and systems-level engineering foundation.